SQL Sales Data Analysis

SQL Server | Data Analyst Portfolio Project

1. Project Overview

This project analyzes sales data using Microsoft SQL Server and SQL Server Management Studio (SSMS) to answer practical business questions related to revenue, customers, products, categories, monthly trends and city performance.

2. Tools & Technologies

• Microsoft SQL Server

• SQL Server Management Studio (SSMS)

• SQL

3. Database Tables

Table	Purpose
Customers	Customer information and city details
Products	Product name, category and price
Orders	Customer orders and order dates
OrderDetails	Products and quantities for each order

4. SQL Concepts Used

• INNER JOIN and LEFT JOIN

• SUM, AVG and COUNT

• GROUP BY and HAVING

• ORDER BY

• CASE WHEN

• COALESCE

• CTE

• RANK() Window Function

• YEAR() and MONTH()

• Subqueries

5. Business Questions Answered

• Total sales revenue

• Top product

• Top customer by spending

• Top revenue category

- Highest-sales month
- Top 3 products
- Most sold product by quantity
- Average Order Value
- Highest-sales city
- Customers spending more than ■50,000

6. Key Business Insights

Metric	Result
Top revenue product	Laptop
Top customer	Rahul — ■172,400
Top revenue category	Electronics — ■274,000
Highest sales month	January 2026 — ■163,500
Highest sales city	Mumbai — ■200,400
Most sold product by quantity	Headphones
Highest quantity category	Accessories — 16 units
Average Order Value	Approximately ■29,979.33
Highest average product price	Electronics — ■36,500

7. Resume-Ready Project Description

SQL Sales Data Analysis | SQL Server

Analyzed sales data to identify revenue trends, top customers, products, categories, city performance and monthly sales using JOINs, CTEs, Window Functions, CASE WHEN, COALESCE, GROUP BY, HAVING and aggregate functions.

8. Project Files

• database_setup.sql — database and table setup

• sales_analysis.sql — business analysis queries

• README.md — GitHub documentation

• Project Report PDF — project summary and insights

9. Conclusion

This project demonstrates practical SQL skills for a Data Analyst role, including relational data handling, multi-table joins, aggregation, advanced SQL querying, ranking and translating query results into business insights.